

Insulin Sensitivity Factor ISF

Contribution to the discussion among DIY loopers

The author assumes no liability

V.3.4.1



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1. What is a ISF?

Your **insulin sensitivity factor** (unit: (mg/dl)/U or (mmol/l)/U) says how sensitive you react to insulin, so how much your glucose level will come down per unit of insulin .

For instance, $ISF = 30 \text{ (mg/dl)/U}$ means, per unit of insulin our glucose level goes down 30 mg/dl. For mmol, divide by 18: $ISF = 30/18 = 1.7 \text{ (mmol/l)/U}$

In Hybrid Closed Loop, the other profile factor, the carb ratio IC, plays the main role around meal times. However, two to three hours after any meal, the administered meal bolus rapidly loses power, but now proteins, fat, and carbs exceeding what could be digested in the first 2 hours come to absorption and require more insulin. This is where your ISF plays the (or a) major role (in iOSLoop, the IC factor continues to play a very strong role).

Remark regarding the workings oforef(1) systems, and the (limited) role carb inputs and the IC value, applied to carb info and/or to observed carb deviation, play in the loop's management of your bg development:

The loop will not take your "e-Carb" inputs for granted and "bolus for it". Your given carb data are used only for one of 4 predictions. The loop also (or *only, if you do no carb inputs*) uses a *prediction of further carb absorption*, which builds on the observed **carb deviation** (=hypothesis, how much got absorbed in the past 5 minute segments). Based on that calculation (which uses ISF), the loop assumes *more* carb decay *in the 5-minute segments that follow*, phasing out the carb decay in the course of the next 1 to max 3 hours. Every 5 minutes this prediction is "met with reality", and the loop further adjusts. This is essentially why oref(1) system allow to go entirely without carb inputs. For more detail on the algo see <https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html#understanding-the-basic-logic-written-version> (or study your SMB tab info). For more info on Full Closed Loop without carb inputs see fcl-e-book in: <https://github.com/bernie4375/FCL-potential-autoISF-research->

Unfortunately, the ISF is not one fixed number you or your loop can count on (count with):

* can vary between times of day => use **circadian pattern**

=> determine it for several times during the day and night, and check whether you see a typical „circadian“ sensitivity pattern corresponding to your bio-rhythm..(As a high basal rate and a low ISF factor are signs of lower insulin sensitivity (=of elevated insulin need), the pattern must be like a mirror-image. More see in section Circadian Pattern, below)

See also: AndroidAPS Users / Files / ..circadian ... Duesterhoff.xls:
<https://www.facebook.com/groups/AndroidAPSUsers/permalink/2869638923257506/>

* will vary between days e.g. when hormones play into it => **Autosens**

* can also fluctuate within shorter episodes, e.g. @ stress, or @ high glucose values because of **insulin resistance** after a fatty meal

* varies also with **TDD** (although, to me at least, it is unclear whether this is an independent effect, or just owed to the fact that on days with high TDD typically more fatty meals are consumed, accompanied by increased insulin resistance) (Plus, we often see “fake insulin resistance” from occlusions, see section 11.)

=> Short time **boosts** may be required, see section 9, further down

=> Based on statistical findings, a **dynamic ISF** formula was proposed that roughly considers most of these effects (see section 3)-

2. Rough estimates for your ISF

a) ... using **Autotune** (not recommended by the author)

Autotune gives you ONE „average“ ISF. Reliability of this result is seen controversially in the loop community, largely, because Autotune relies on your data inputs which probably contain errors (like your carb inputs). Also there is no use to Autotune when you are using dynamic ISF adaptations (dynamicISF, autoISF).

See also: <https://androidaps.readthedocs.io/en/latest/Usage/autotune.html>

b) ... via **daily total insulin dose, TDD**

The daily total insulin need describes, roughly, the insulin sensitivity of a type1 diabetic.

The following table reflects values seen in a big study.

Example: At 35 – 40 units of daily insulin (mid of left column) an ISF of around 45 mg/dl/U (35-60) would be suspected, or around 2.6 mmol/l/U

Estimating the Sensitivity Factor based on Total Daily Insulin

Average Total Daily Insulin (all basal + all boluses)	Sensitivity Factor (mg/dl) how much 1 unit lowers blood sugar	Sensitivity Factor (mmol) how much 1 unit lowers blood sugar
5 units	320-360	18-20
7 units	220-260	12-14
10 units	155-185	8.6-10.3
12 units	125-155	6.9-8.6
15 units	95-125	5.3-6.9
18 units	80-110	4.4-6.1
20 units	70-100	3.9-5.5
25 units	60-80	3.3-4.4
30 units	50-70	2.8-3.9
35 units	40-60	2.2-3.3
40 units	35-50	2.0-2.8
45 units	30-45	1.7-2.5
50 units	30-40	1.7-2.2
60 units	25-35	1.4-2.0
70 units	20-30	1.1-1.7
80 units	20-25	1.1-1.4
100 units	15-20	0.8-1.1
120 units	13-17	0.7-1.0
140 units	11-15	0.6-0.8
160 units	10-12	0.5-0.7
180 units	9-11	0.5-0.6
200 units	8-10	0.4-0.6

c) Estimate using the „1700“ rule

According to a study <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4454102/> the ISF can be estimated using:

$$ISF = 1700 / TDD \quad (\text{or} = 94.4 / TDD \text{ for mmol})$$

where *TDD* is the total daily insulin, determined for each virtual patient, and assuming an average diet of 180 g of carbs for adolescents and adults, and 135 g for children.

Example as above, 35...40 U TDD: $ISF = 49 \dots 42 \text{ mg/dl/U}$ (or divided by 18 for mmol)

3. Dynamic ISF

3.1 Using a dynamicISF derived from bg value and TDD: The 277,700 rule

It has been suggested

- <https://www.facebook.com/groups/TheLoopedGroup/posts/2996545667228746/> and <https://youtu.be/oL49FhOts3c/> Chris Wilson / see also in Loop&LearnFB)

that the ISFs can be dynamically modelled using the formula:

$$ISF = 277\,700 / (TDD * bg) \quad (or = 15\,428 / (TDD * bg) \text{ for mmol})$$

It would be super convenient if that works well enough, because it would not require the user to determine his ISF factors at all.

Discussion: The formula is derived from real-life data of non-looping insulin-dependent diabetics. Loopers would be a drastic outlier in the spectrum of (high) bg values, and resulting HbA1c's seen, in that group.

To make ISF smaller (=more aggressive) at higher daily insulin use (TDD) *in prior days* .. 8 hours might be useful if your lifestyle is un-steady in this respect. However, we easy can imagine situations where this TDD concept would backfire:

- e.g. after the decision to eat less because the last days were too calorie rich
- e.g. if TDD was rising recently because an old pod increasingly went towards occlusion, but now starting a 100% delivering new pod.

Moreover, when looping, it is not ideal to be more aggressive when bg is very high. Ideally, at bg peak the loop zero-temps because it received sufficient insulin “on the way up”. (And, to receive sufficient insulin then, i.e. already at lower, but rising, bg values, an aggressive ISF also at lower bg is helpful).

To resolve some of these issues, the dynamicISF we now find integrated in looping systems was conceptually refined.

3.2 Refined dynamicISF variants

Plug-ins are available for both, AndroidAPS (by Tim Street), and also of Trio and iAPS. To overcome major problems with the rough first approach (above ISF formula, to put it bluntly, was based on “wild” data from how “real average diabetics jerk around with their super high glucose values”), the formula has been significantly refined, and also was made user-tuneable.

Key improvements were

1) to allow to shorten the used observation period for TDD (towards the magnitude used also in Autosens; see also: <https://androidaps.readthedocs.io/en/latest/DailyLifeWithAaps/DynamicISF.html#enable-tdd-based-sensitivity-ratio-for-basal-and-glucose-target-modification>)

2) not just use bg, but (also, in various degrees) strongly consider predicted bg (details for AAPS see: <https://androidaps.readthedocs.io/en/latest/DailyLifeWithAaps/DynamicISF.html#future-isf>) and for Trio: see here: <https://open-iaps-docs.readthedocs.io/en/latest/settings/configuration/concepts/autosens-dynamic.html#advanced-information> and here: <https://diy-trio.org/videos/v/everything-to-sell-anything-427qe-t8t94-vs9pb-p8b9h>)

3) most (all?) variants come with a user setting (in AAPS called “dynamicISF adjustment factor% (1...300%) that multiply with the divisor in the dynamicISF formula. For example: 1/ 200% halves the dynamicISF = makes it double as aggressive, while 67% would in effect only deliver 2/3 of the insulin).

4) some variants (not the AAPS one, afaik) do no longer fully de-couple from the user’s profileISF, but come with a user setting that influences how strong the dynamicISF can deviate from profileISF.

Using the predicted bg is a great idea, but it comes with two very principal problems:

1) There is no predicted bg, but there are 3 or 4 different prediction curves, and just in the most critical decision times when we need more insulin, they tend to diverge very much.

- (They narrow down while the loop usually already zero-temps. Then, we really do not need an ISF any more).

So, it is a tough call, how many minutes out, and on (or between) which of the prediction curves, the “predicted bg” shall be located, for use to set the dynamic ISF for the next dosing decision.

- This is a general problem *for all loops* regarding determining SMB sizes. To safeguard from hypos, they all “pick” one of the lower predicted bg values, to base their dosing decision on. But *dynamicISF* additionally brings this uncertainty into the decision, which ISF to use.

2) Evidently, an ISF must be used to calculate all the prediction values. So, then to use those to define the ISF is a bit unlogical - but it is the only thing available for, hopefully, a good-enough approximation.

Still, dynamicISF may be the easiest way (with just 1 tuneable factor) to arrive at ISFs that by and large work well for you.

3.3 Alternatives to using dynamicISF

Profile_ISF + Automations

This paper focusses on *kind of a bottom-up personalized approach* to good profile_ISF settings via:

- Experimental ISF determinations (see next section 4.) leading to a circadian 24 h pattern of ISFs

- 151 - use of Autosens
- 152 - boosting ISFs at high glucose values, notably if owed to post-meal insulin resistance
- 153 (see section 9.)

154 **AutoISF**

155 autoISF is also anchored on experimentally determined profileISFs.

156 The method is sketched in section 10, and better described in section 4.1.2 of: "04_Meals.ISF-
157 weights....pdf" (in Github/bernie4375 "FCL...autoISF..." branch)

158 If done right, one of the *bottom-up* approaches, based on your personal data, should be
159 better than dynamicISF, and also more real-time reactive to changing insulin requirements,
160 notably in full closed looping.

161 Please apply caution if you *combine* some of the discussed methods, as they probably work
162 in the same direction, and *could „over-adjust“* your ISFs.

163

164 **4. Determination of Carb Rise Ratio CRR and Insulin-Sensitivity Factor ISF**

165 **Basal rate first**

166 Verify that you have a correct basal rate before determining factors!

167 Else your ISF will have to balance errors in your hourly basal pattern, and you may
168 lose some of the principal power of your loop.

169 Helpful links regarding basal rate

- 170 • A basal rate helper, drawing data from your Nightscout site (by Peter van Rijt):
171 <https://www.nightscoutsuggestions.com/index.php>
- 172 • a good article in support of that process: [https://www.mysugr.com/en/blog/basal-rate-](https://www.mysugr.com/en/blog/basal-rate-testing/)
173 [testing/](https://www.mysugr.com/en/blog/basal-rate-testing/)
- 174 • a process description by a commercial service, including a tool :
175 <https://www.mevita.de/.../online-zugang-erstellung-einer.../>

176 Although I advise to determine your true, likely **circadian**, basal rate: It can be „good
177 enough“, and advantageous especially for small kids, to simplify things and assume a (in
178 case of doubt, low) **flat basal rate**, together with a well-contoured ISF pattern („with enough
179 bite where it counts“). Discussions on this topic can be found in looper sites using the search
180 term “flat basal rate”. Interpret enthusiastic reports on flat basal with caution, though. Positive

181 observed effects are likely coming from significantly increased attention to ISF ... and this
182 brings us back to our topic☺ ...

183 **Start your ISF test with a CRR determination**

184 To determine ISF, select a day without prior strenuous exercise/activity, without exceptional
185 stress, or infection.

186 As testing for ISF must start out with a high plateau of glucose level, it makes a lot of sense
187 to do a determination of your **CarbRiseRatio CRR** first:

188 (In a relatively stable, normo-glycemic phase) take a sweet drink with a known carb content,
189 and watch (in Open Loop, not giving any insulin except profile basal rate) by how many
190 (mg/dl) glucose rises, until reaching a plateau (in about 1 hour).

191

192 *Example: After consuming 20g, my glucose rises from 90 to 190 mg/dl.*

193 $CRR = (190-90)\text{mg/dl} / 20\text{g} = 5 \text{ mg/dl} / \text{g}$

194 **ISF determination**

195 Now, having reached a high plateau – *at 190 mg/dl in our example* ,-we move on to
196 determine the **ISF**.

197 Inject (still in Open Loop, with profile basal running), an amount of insulin as you typically
198 would (considering your personal insulin sensitivity). *Example: 2 units*

199 **Caution:** For a kid you would use much less than 2 U: Think about typical correction
200 boli in the past, or use your “suspected” ISF (if it is over 100 mg/dl/U then you
201 definitely should stay under 1.0 U)..

202 Then watch for about three hours until a deeper plateau is reached, *for example at 110*
203 *mg/dl*. Then you can evaluate:

204 *Example: 2 U of insulin lowered my glucose by 80 mg/dl . Hence, ISF = 40 mg/dl / U*

205

206 *Off topic remark: Using CRR and ISF you can also calculate your IC for this time of day, or*
207 *check the **plausibility of the IC** you are already using, because:*

208 **ISF** (mg/dl decrease per unit of insulin) / **CRR** (mg/dl increase per g of carb) = **IC** (g
209 carb per unit of insulin) (similar in mmol) or also: $CRR = ISF / IC$

210 *IC in the example from above: $ISF / CRR = 40 \text{ (mg/dl)/U} / 5 \text{ (mg/dl)/g} = 8 \text{ g/U} = IC$*

211 See also:: <http://seemycgm.com/2017/10/29/fine-tuning-settings/>

212 The CRR is relatively stable (does not vary by much over 24 hours):

- 213 - while both ISF and IC do vary (in similar ways, related to your insulin sensitivity)
- 214 - if you divide one by the other, deviations owed to sensitivity cancel out.

215

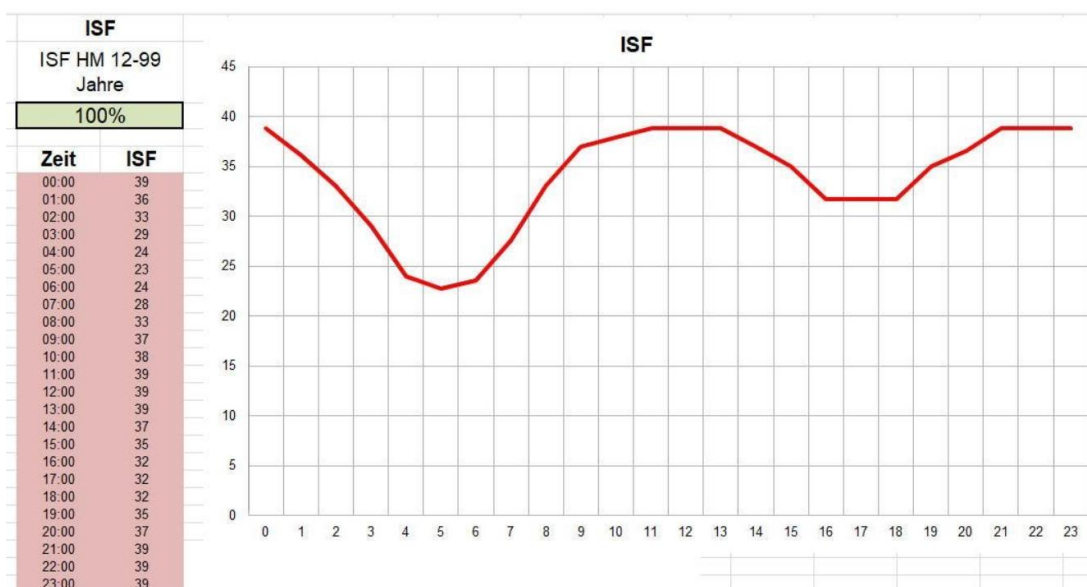
216 5. Circadian pattern of ISF

217 ISF varies over 24 hours according to a „circadian“ pattern of varying sensitivity to insulin.

218 See AndroidAPS Users / Files / ..circadian ... Duesterhöft.xls:

219 <https://www.facebook.com/groups/AndroidAPSUsers/permalink/2869638923257506/>

220 Example



221

222 Do not just copy a pattern that may be valid on average for T1Ds out there, or for a single
223 other person! For instance, I am more of a night owl, so that curve shifts 1-2 hours to the
224 right. Likewise, if you do shift work or cross-time-zone travel, keep an eye on your (probably
225 time-delayed) adjustments of your body's insulin sensitivity.

226 The deepest dip in ISF, and highest hill in hourly basal, is typically an hour or two before
227 getting up and breakfast. However, I usually skip breakfast, so I „break my fasting“ (!) for
228 lunch and see higher insulin sensitivity there than the standard curve suggests. (Another
229 explanation could be, that others would have tail of insulin sensitivity from breakfast still at
230 work supporting insulin for lunch (with a typical breakfast, low in fat and protein).

231 *In case you work with a flat basal rate, your ISF should look even more „contoured“:*

232 For instance, if you use your low hourly basal, as appropriate in most night hours, for all
233 24 hours, the daytime ISFs would need to be make up for „deficiencies“ in your hourly
234 basal; likely morning and evening ISFs would dip relatively lower even than in picture
235 on preceding page. YDMV!

236 So, determine **your** ISF for a couple of times, in which you find high values which your loop
237 needs to regulate down (one of them, inconveniently, probably being after midnight; see also
238 2 sections lower).

239 You can interpolate for the hours for which you did *not* „experimentally determine“ your ISF.
240 Follow the idea (or data) you have regarding your general pattern of insulin sensitivity.

241

242

243 **Caution regarding kids**

244 For kids, the 24 hour sensitivity patterns can differ strongly from adults and can also change
245 strongly in certain growth phases, see the following graph (shared by „basal rate guru“ Heiko
246 Mueller, www.mevita.com)

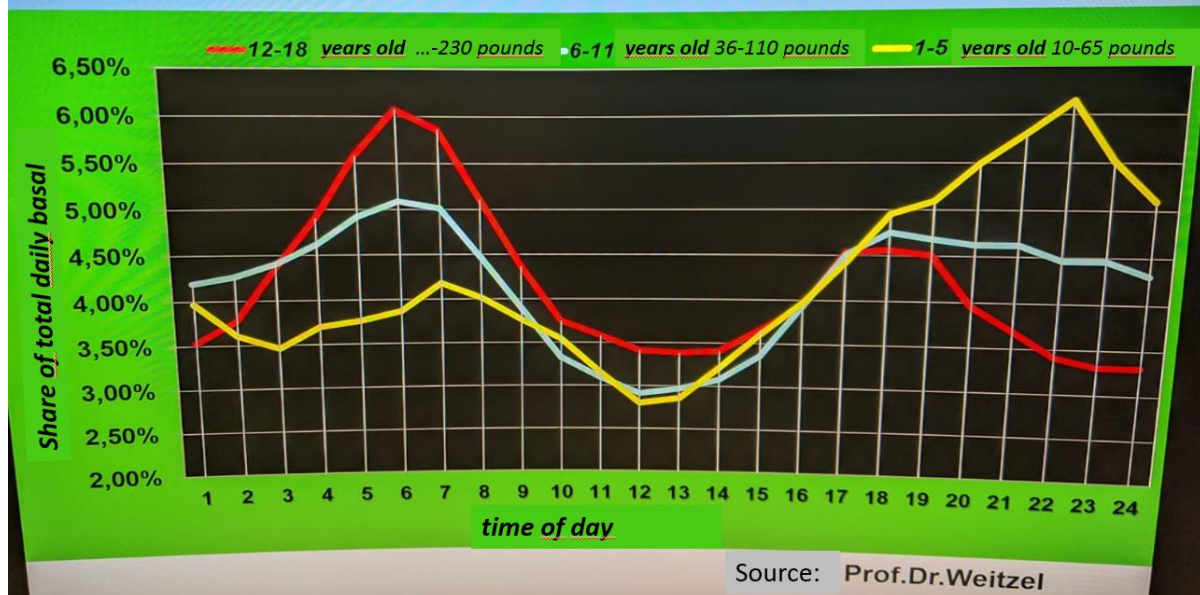
247 To help “construct” a circadian profile for kids, there is a “profile helper” in AAPS:
248 [https://androidaps.readthedocs.io/en/latest/Configuration/profilehelper.html#profile-](https://androidaps.readthedocs.io/en/latest/Configuration/profilehelper.html#profile-for-kids-up-to-18-years)
249 [for-kids-up-to-18-years](https://androidaps.readthedocs.io/en/latest/Configuration/profilehelper.html#profile-for-kids-up-to-18-years)

250

251 The following graph shows typical circadian patterns for basal rates, and is normalized to
252 100% being the total daily basal.

253 So, the graph below says how much %of the daily total is needed in every hour of the
254 day. It does NOT show the very significant differences in basal insulin need between
255 the age groups! We are looking only on the 24 hr patterns here!

Basal rate (hourly as % of daily total) in children and youth, during the day (might shift horizontally if untypical bedtimes)



The graph is for basal pattern.

For ISF the curve would look „opposite“ (~mirrored on x-axis), because **higher** basal („mountain“) = lower insulin sensitivity = higher insulin need = **lower** ISF factor („valley“). For the red curve (age 12-18 resembles adults fairly well already) we see this mirrored pattern beautifully in the ISF graph, 2 pictures higher up.

6. Signs for a wrong ISF when looping

If your ISF is too strong (in other words the actual number is too low compared to reality of your real insulin sensitivity), one of the most common symptoms you'll see in closed-looping is a roller coaster of glucose values, and the temporary basals might be cycling between zero and high temping.

If ISF is too weak, the glucose is not corrected down to target (or this happens only extremely slow).

If you observe such behaviour in certain time periods of your day, you might consider changing the ISF by plus 10-20% (if ISF seemed too strong) or by minus 10-20% (if ISF seemed too weak), and see (with alarms set on loud), whether the situation improves next day.

7. Tuning your profile ISFs in Closed Loop

Caution: The following applies to using and tuning profile ISF!

Tuning *dynamic ISF* (if you use that plug-in) requires different considerations:

Look up in your data which part of the “dynamic range” of ISFs resulting from the used formula is actually in use to correct your glucoses, and how to influence that via

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280 The effects of a suboptimal ISF settings as discussed in the preceding chapter can be seen
281 nicely in the night following a bigger dinner. When the meal time bolus loses power, about 2-
282 3 hours after starting the meal, often relatively high glucose values exist at bedtime for the
283 loop to take care of.

284 With a good ISF setting, the glucose curve will, within a few hours (while you are sleeping),
285 go down and „hug“ the glucose target you have set for the night time.

286 If the glucose curve swings very much below target, or even triggers a hypo alarm, ISF is
287 probably too strong, and you should elevate the nighttime values in your profile and observe
288 again in the following nights.

289 Conversely, if the glucose hovers for hours above target: In that case you would try with
290 lower ISF next nights. Make changes in small steps (hypo danger) . Don't do this after a day
291 with strong activity/exercise. Setting the nighttime glucose target a bit higher for the time of
292 testing can also be a good idea.

293

294 8. Avoiding hypos after correcting high glucose values

295 The course of insulin activity during the 5 hour+ „DIA“ period of any administered bolus
296 implies that anything you try to achieve for short-term glucose control (1-2 hour time frame),
297 comes with a „burden“ of effects that tend to drive your glucose below target a bit later

298 A high glucose value therefore can not be corrected rapidly without running a risk to go into a
299 hypoglycemia 2 or more hours later (as probably everybody has experienced with impatient
300 „rage“ bolusses).

301 All loops apply caution in that they analyze the situation every 5 minutes, and generally apply
302 much less than the calculated InsulinRequired (AndroidAPS generally applies 50% of what
303 the calculation suggests).

304 (To avoid misunderstanding: This does not mean that 50% of insulin is really withheld ; what
305 was not given in the preceding 5 minute segment, is likely missing „on top“ also in the
306 following 5 minutes, and gets delivered to, again, 50% then , now making up already 75% of
307 the previous InsulinRequ. ...).

308 Users of iAPS and of AAPS dev variants that allow SMB delivery ratios bigger than
309 0.5 should be aware that changing to 0.6 means already a 20%, and to 0.75 a 50%
310 boost for the insulinRequired calculation. Going far over 0.5 can become dangerous,
311 e.g. if a single jittery CGM value can be trigger much bigger SMB than really needed.

312 It is important to watch your loop with some patience, and to analyze what (if anything) you
313 might try differently *for the next day(s)*.

314 In case your loop is acting too aggressively, you may need to take a couple of grams of
315 carbs to avoid going low. (If you use AndroidAPS, your main screen probably shows carbs
316 required in next minutes, to boost the cob number which is given right next to it), Note down
317 how many grams you took *for example*,. *4 grams* . Using your IC factor, e.g.. *8 g/U*, calculate
318 how much insulin (*0,5 U*) would be needed for absorbing these extra grams. Your post-
319 prandial SMBs should have given that amount of insulin **less** (assuming you observe that
320 things go well with these extra 4 grams) For example, if your pp. SMBs were *4 U* in total, *you*
321 *should try 3.5 U next time.. Your ISF factors in the relevant hours should be elevated by 14%*
322 *(=4 / 3.5)*.

323 To summarize:

- 324 1. Try to avoid high glucose values (see material on meal management in hybrid closed
325 loop)
- 326 2. If high values occur: First exclude occlusion as underlying problem. (see section 11).
- 327 3. Watch and analyze your loop preferably without giving extra boli (but take, and
328 record, anti-hypo snacks if needed), Define what you might change to improve in the
329 next days.
- 330 4. In case you think you see very short-lasting resistance, e.g. from fatty meals, consider
331 brief boost of ISF (next section).

332

333 9. **Short-time „boost“ of ISF** (in AndroidAPS also via **Automation**)

334 Background: Often a short phase of insulin resistance is encountered a few hours after a
335 fatty meal. Insulin seems to have lost it's potency for 1 or 2 hours. This is often assigned to
336 receptors blocked by free fatty acids, and requires extra insulin – but only for a very limited
337 time, until the receptors are cleared -.

338 Rather than taking care of this phenomenon by giving an extra manual bolus, AndroidAPS
339 allows an elegant automated solution for this problem: When conditions like glucose above
340 180 mg/dl (and others, like iob thresholds, time of day where the problem is likely etc) are

341 fulfilled, a more aggressive ISF can be selected (*for example, by setting 130% profile*
342 *(,override') for 8 minutes.*

343 Such automations are technically very easy to „program“, but require attention to one's own
344 daily patterns, to define a truly personalized problem solution.

345 Details see for example at

346 <https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#stagnation-at-high->
347 [bg-values](https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#stagnation-at-high-)

348

10. autoISF

autoISF is a feature (currently, Jan. 2025, not yet included in AndroidAPS Master), which allows **flexible boosting** of ISF. (So it will *not* help you “find your ISF values”, like dynamicISF promises to).

autoISF **uses your profile ISFs**, to adjust them further for optimal management of how your glucose curve is predicted to further develop.

The working principle of Auto-ISF is, to provide sharpened ISF factors (and there are various new parameters in /Settings to tune this individually), depending on;

1) Glucose plateau = how high glucose is, and for how long it is that high already. – (This provides a “more elegant” solution of the problem we just discussed in section 9).

2) Glucose level = allows to sharpen ISF at high glucose levels (I am not using this, because it is kind of late to react then; also, if my glucose runs really high, this is usually occlusion related = ISF does not help at all then, see section 11. But in Hybrid Closed Loop, if you often find yourself high when your given bolus expires, the feature might help/YDMV)

3) Glucose delta = sharpen ISF when glucose rises strongly. It makes a lot of sense to act BEFORE glucose is high, and you can tune aggressiveness according to personal needs.

4) Acceleration of glucose rise = enables strongest ISF as soon as a meal-related rise is detected (and, in balance, allows reducing the aggressiveness – already during the glucose rise! - when curve bends around heading towards a peak). This is extremely helpful in UAM Full Closed Looping, but requires a smoothly operating CGM, so what the sophisticated math makes of your glucose curve, to fire off boosted SMBs, is not just a temporary artefact.

Any of these 4 autoISF varieties can (must) be tuned separately, but also can interfere with each other and with Autosens. For more details see fcl-e-book, chapter 4.1.2 in:

https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/04_Meals.ISF-weights_FCL-book_V%204.7.pdf

11. „Differential-diagnosis“ occlusion when dealing with high glucose values

Back to easier stuff, relevant to us all.

When we encounter high glucose values it is very **important**, as a first step, **to exclude a compromised insulin delivery** as underlying cause.

Users of Lyumjev or of Fiasp seem to experience occlusions quite often, especially if not changing cannula frequently. Under the skin, near the insulin insertion site, partial or full blockages can occur and insulin is being pushed back out into the adhesive patch of the cannula. This results in glucose rising despite **(fake) high iob**.

In such instances, loop should be switched off. The share of insulin that did not effectively enter the body system must be estimated (conservatively, in case of doubt), and erased (in AAPS: Treatments tab/Bolus/erase a number of entries there), so a realistic iob number is there for the loop to continue. The occlusion must be resolved (usually by setting a new cannula), and the high glucose must be treated. As your iob estimate is likely uncertain, do this conservatively, and continue in closed loop with alarms set on loud.

In case the high values were realized only after a couple of hours, watch out for signs of **Keto-Acidosis** and be ready to follow a treatment scheme for it. Hospitalization may be required.

Off Topic: Also **fake cob** can happen. This would develop **after throwing up** shortly after a meal which you bolussed for. As insulin cannot be removed from the body, and glucose snacks etc.- which are needed in significant quantity quickly now -, likely are also thrown up, a glucagon injection may be needed.

Caution: To stop more insulin delivery, and erase false cob, is by far not sufficient to resolve this crisis. This is an **emergency** with extremely high hypo danger at the rapidly approaching point of time, when your meal bolus reaches maximum activity! Call for emergency:

you can't eat/drink (or not keep it in)

* if you go below a certain glucose value that you laid out for yourself when still clear in the head. (It probably would be too complicated to build a strategy on the 5-minute delta glucose values, and amount of carbs you were successfully taking)

Especially users of Lyumjev or Fiasp should be aware how fast glucose can sink in absence of sufficient carbs for the iob that is present.!

12. Remarks regarding „UAM“ Full Closed Loop

With well-functioning CGM and a rapid insulin (preferably Lyumjev), more and more – currently mostly adult – loopers establish a reasonably functioning Full Closed Loop with AndroidAPS (also without having a low carb diet). More info see here:

<https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html>

In Full Closed Loop (UAM/no bolus/no carb inputs) **ISF is the key parameter because** no more meal boli are given, and **the loop exclusively reacts on rising glucose**.

Full Closed Looping had been pioneered by some low carb OpenAPS Loopers, but picked up momentum when in 2020 Lyumjev became available in an increasing number of markets.

There are currently several avenues established or explored towards a no-bolus no carb-inputs Full UAM Closed Loop. For more info see fcl-e-book in:

<https://github.com/bernie4375/FCL-potential-autoISF-research->